UB HACKING 2026 · DEEP-SPACE SAMPLE BAY

STARHAND

Reach into a deep-space sample bay with your bare hand.

Ahmed Sohail Butt
UNIVERSITY AT BUFFALO · NOV 7–8, 2026



BENNU · NASA SHAPE MODEL · 31 MM · 15 G

Nothing may touch Bennu

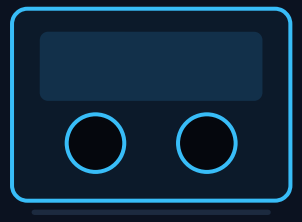
121.6 g

of asteroid Bennu, returned to Earth by NASA's OSIRIS-REx

INSIDE THE SAMPLE

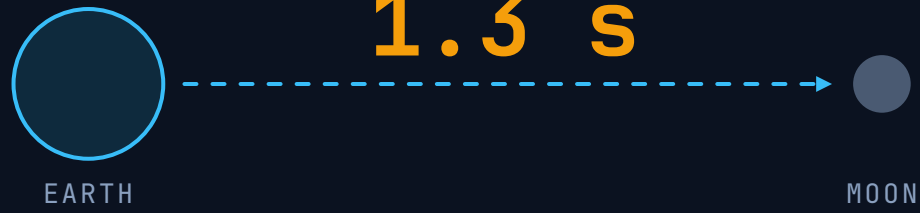
5 of 5
nucleobases

14 of 20
amino acids life uses



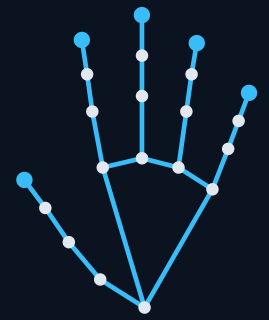
SEALED GLOVEBOXES

NASA Johnson Space Center. Only clean tools touch it.



One-way light time. Remote hands need safe autonomy.

Sense → Understand → Act

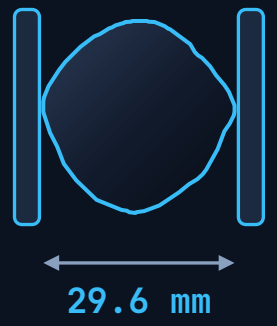


01 SENSE

Hand tracking, on-device

Fixed bay camera

~30 HZ · 21 LANDMARKS



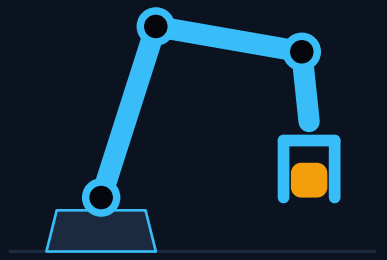
02 UNDERSTAND

Gemini designs samples

Code measures grip, height, mass

Robotics-ER checks the bay

ONE UNIFORM SCALE, OR REFUSE



03 ACT

SO-101 follows your hand

Pinch to pick, pinch to place

REACH 130–200 MM · HOVER 60 MM

Live demo

SCREENSHOT PLACEHOLDER

screenshots/bay.png

Capture the bay view, ideally at 16:9.
Save it at this path, then re-run build_deck.py.

Gemini proposes, code decides

GEMINI

Designs a 3D shape program

```
{"parts": […]}
```

TRUST BOUNDARY

CODE build → measure → guardrail

grip	15–45 mm
height	≤ 45 mm
mass	≤ 100 g
footprint	≤ 60 mm

SCALE TO FIT



REFUSE

and says why, out loud

MEASURED IN THE APP · SO-101 LIMITS

✓ **Iron meteorite** 38 mm, 204 g → too heavy → ×0.76 → **28.9 mm grip, 19 mm tall, 90 g**

× **Antenna boom** refused “too tall and thin… make it at least 40 mm across, or no taller than 11 mm”

Hand signals beat the AI

WHO WINS, HIGHEST FIRST

HAND SIGNALS › PANEL › CODE GATE › AI

STOP
Open palm
SPEED → 0

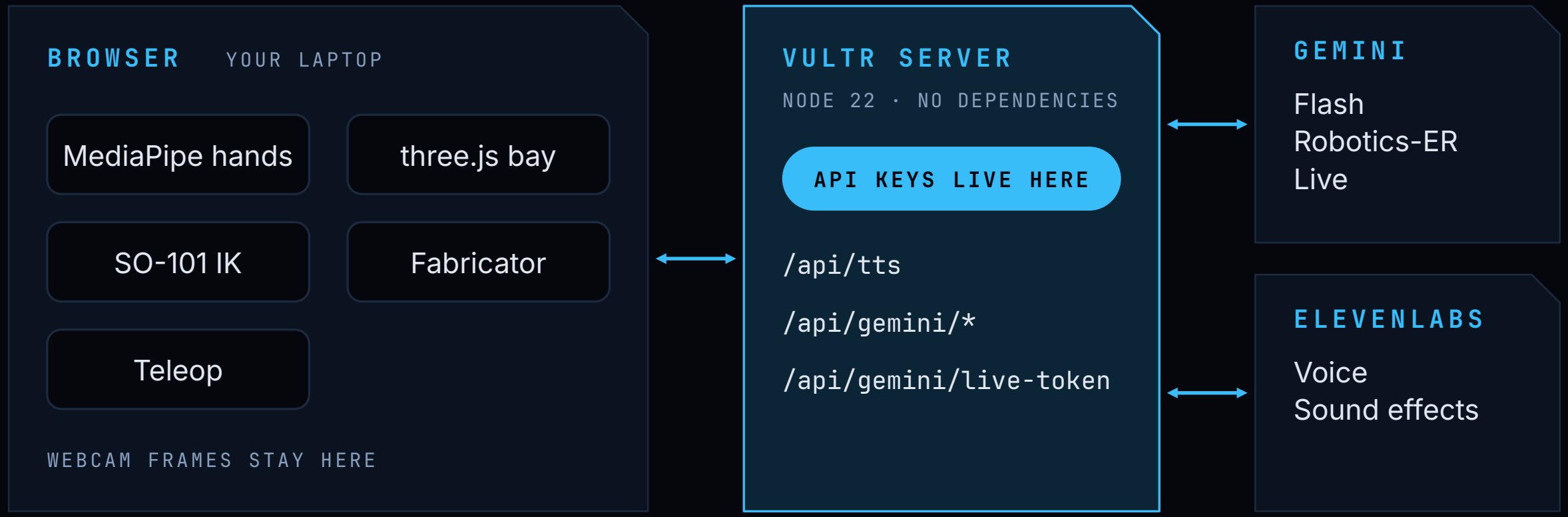
E-STOP
Held fist
TELEOP OFF

RESUME
Thumbs-up
BACK TO RUNNING

HOLD
No hand in view
DEAD-MAN: ARM WAITS

Gemini can **never** start, stop, resume or reset the bay.

No API key ever reaches the browser



Hand tracking runs on your device. Gemini Live gets a short-lived token, never the key.

What's real, what's simulated

REAL

- Hand tracking: webcam, on-device
- Voice: Gemini Live in, ElevenLabs out
- Gemini and ElevenLabs API calls
- SO-101 geometry: MuJoCo Menagerie
- NASA asteroid shape models



BENNU

ITOKAWA



EROS



KLEOPATRA

KINEMATICS CROSS-CHECKED VS MUJOCO: < 4e-8 m

SIMULATED

- Arm: kinematic, no physics
- Grasp: attaches by rule
- Mass: assumed densities

DISCLOSED IN THE APP AND README

CREDITS SO-101 model: MuJoCo Menagerie (Apache-2.0) · Asteroids and Mars: NASA 3D Resources, Itokawa NASA/JAXA, no endorsement implied · three.js, React, MediaPipe, @google/genai

AI TOOLS Claude Code wrote most of the code under my direction · Gemini and ElevenLabs run in the app · all code written at UB Hacking

What we learned, what's next

NEW SKILLS

01 On-device hand teleop

02 AI shape programs + measured guardrails

03 ElevenLabs voice design

04 Deploying on Vultr

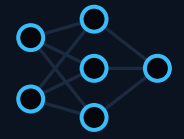
NEXT

1 Record teleop sessions as LeRobot datasets



2 Train ACT policies

ZHAO ET AL., 2023



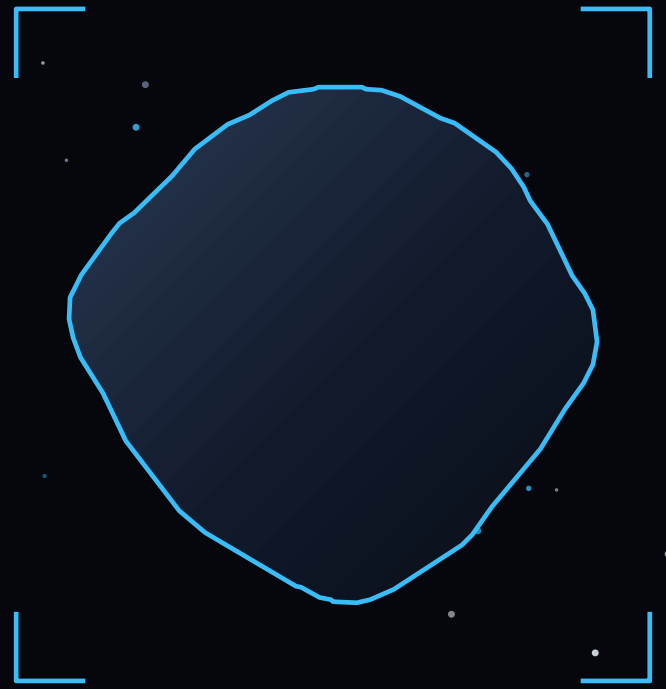
3 Run it on a real SO-101 arm



STARHAND

Thank you

Reach into deep space, safely.



REPO github.com/ahmedsohail2003/buffalo-hackathon

LIVE <live URL on Vultr>

VIDEO <YouTube link>

Ahmed Sohail Butt UB HACKING 2026

Teleop

SCREENSHOT PLACEHOLDER

screenshots/teleop.png

Capture the hand pane next to the bay, ideally at 16:9.
Save it at this path, then re-run build_deck.py.

Your hand sets where. A pinch says when. No hand: the arm holds.

Fabricator

SCREENSHOT PLACEHOLDER

screenshots/fabricator.png

Capture a Fabricator result card, ideally at 16:9.
Save it at this path, then re-run build_deck.py.

Gemini designs it. Code measures it, then scales it or refuses.